Lab 10: set up a closed GSE vessel

Guide page 1 | Worksheet D, p. 3 | Predict at slide 15, watch slide 16, rerun at slide 18.

Open Lab 10: Solid equilibria

The lesson uses the GSE part of Lab 10. The SLE equation is a hand calculation; the binary eutectic interface is optional. Select **Experiment → GSE · sublimation in a rigid vessel** first, because changing mode resets its synthetic defaults.

Field	Enter / check
Property sources	Synthetic teaching definitions; not experimental properties.
Temperature / K	300
Minimum / maximum valid T / K	250 / 350
Subliming component / inert carrier	Synthetic volatile solid / Synthetic inert gas
Fixed free gas volume / L	1
Total volatile inventory / mol	0.010 for case 10A
Inert gas inventory / mol	0
Reference temperature / K	300
Sublimation pressure at reference / kPa	10
Sublimation enthalpy / J mol^{-1}	50000

Click **Calculate equilibrium →**. Read the phase label and results table. At T=Tref, the supplied sublimation pressure equals Pref. The entered ΔHsub controls the temperature sweep; it does not change Psub at that reference point.

Before changing anything

Record PA, total pressure, volatile gas amount, remaining solid and the balance residual. Use ncap=PsubV/(RT) to check the result independently. Name your run 10A in your worksheet and, if desired, in the workspace field **Label for the next captured case**.

Select **Export study JSON**. The file downloads and is also captured in the in-memory workspace. The worksheet is your explanation; the exported study is a reproducible input/result record. Neither is automatically submitted.

Lab 10: compare, save and reopen

Guide page 2 | Three controlled cases | Keep the same T, volume, property relation and declared range.

Case	Change from previous case	Record
10A	Start nA=0.010 mol, inert=0	Phase, PA, Ptotal, gas and solid
10B	Only nA → 0.002 mol	Same quantities; explain depletion
10C	Only inert → 0.002 mol	Separate partial from total pressure

After every input change, click **Calculate equilibrium →** before recording or exporting. The interface clears old results when inputs change. A disabled export button or “Inputs changed” message means the current inputs have no fresh result.

Read both plots

Choose **Partial and total pressure**, then **Gas and solid amounts**. The horizontal axis is temperature across the declared range, not time. The selected 300 K result is in the table. Compare actual volatile partial pressure with sublimation pressure; total pressure also includes inert gas. In the amount plot, gas plus solid is the volatile inventory and excludes inert gas.

Where solid remains, PA=Psub. After the solid inventory is exhausted, PA can be lower. A negative $(\mu_{gas}-\mu_{solid})/RT$ then refers to the hypothetical absent solid and is compatible with stable all-vapor equilibrium. Gaps can indicate a pressure guard; do not interpret a missing segment as zero.

Save and reopen an actual calculation

1. Export each fresh case. Open **Study workspace · reopen and compare cases** near the top of the page. Check the captured case labels.

2. Click **Download comparison**. Keep the JSON with your worksheet. It contains records from this Lab only; Lab 11 uses its own comparison file.

3. To verify a saved file, use **Load studies or a comparison JSON**. If necessary, download first and use **Clear comparison** to avoid duplicate cases or the eight-case limit. Clearing the in-memory list does not delete downloaded files.

4. Click **Restore** beside the intended case. Verify the mode, temperature, volume, inventories and property values. Click **Calculate equilibrium →** again. Compare the fresh result with the recorded values; imported results are evidence, not a fresh solve.

Raw JSON units: Lab 10 stores pressure fields in Pa and volume in m^3: input.Pref=10000 means 10 kPa; input.volume=0.001 means 1 L. Workspace controls preserve the values entered in the labeled interface. Do not treat raw-field numbers as display units.

Keep one annotated plot, three records and an independent balance. Model assumptions: ideal gases, one pure solid, no adsorption or reaction, fixed free gas volume, and no liquid formation in the declared range.

Lab 11: compare uptake models

Guide page 3 | Worksheet E-F, p. 4; G, p. 5 | Predict at slide 29, watch slide 30, rerun at slide 32.

Open Lab 11: Adsorption equilibrium

Work in **Explore isotherms**. Select **Reset synthetic parameters** if you need the defined starting point. The experimental fitting section below is optional and is not required for this lesson.

Field	Value
Adsorbate / adsorbent	Synthetic gas / Synthetic adsorbent
Parameter source	Synthetic definitions at 300 K
Parameter and vessel temperature / K	300
Minimum / maximum model pressure / kPa	0 / 100
Model to edit and use in the vessel	Langmuir initially
Capacity qs / mol kg^{-1}	2
Affinity b / kPa^{-1}	0.1
Henry K / mol kg^{-1} kPa^{-1}	0.2 (select Henry to inspect/edit)
Closed vessel: free gas volume / L	1
Dry adsorbent mass / g	10 initially
Total adsorbate inventory / mol	0.010
Nonadsorbing inert gas / mol	0

Expand **Closed vessel** to inspect its inputs. Return the selected model to Langmuir and click **Compare & solve vessel →**. The isotherm plot shows all four independently specified models, but the vessel table uses only the selected model.

Pressure is adsorbate partial pressure in kPa. Uptake is absolute mol/kg of dry adsorbent, numerically equal to mmol/g. The interface takes mass in g and volume in L; use kg in your hand balance. Temperature labels supplied parameters; changing T alone does not predict new adsorption parameters.

Lab 11: balance and retain evidence

Guide page 4 | Finish G, then submit I; H is optional | No experimental data upload or fitting is required.

Case	Selected model	Dry mass / g	Other inputs
11A	Langmuir	10	Fixed as guide p. 3
11B	Henry	10	Only model changes
11C	Langmuir	20	Compare with 11A

For each case, click **Compare & solve vessel →**, record PA, q, gas adsorbate amount and adsorbed amount, and click **Export model study**. Input changes invalidate the previous result. Save each completed case before changing a control.

Calculate the independent balance

$$n_A = P_A V_g/(RT) + m_s q(P_A)$$

Use R=8.314462618 kPa L mol^{-1} K^{-1}, Vg=1 L and ms=0.010 or 0.020 kg. With no inert gas, total gas pressure equals PA. Check the separate gas and adsorbed amounts, not only the small residual reported by the solver. Rounded display values can create a small hand residual; JSON preserves additional digits.

Matching KH=qs b gives the same zero-pressure slope. It does not make Henry and Langmuir equivalent at finite pressure. For a larger adsorbent mass at fixed total inventory, distinguish lower uptake per kg from the total amount adsorbed on all of the solid.

Reopen and recalculate

Open **Study workspace · reopen and compare cases**. Download the Lab 11 comparison, then load it through **Load studies or a comparison JSON**. Restore a chosen case, check the selected model and all vessel inputs, and click **Compare & solve vessel →** for a fresh solve. Keep the comparison and worksheet together. Do not mix Lab 10 and Lab 11 files in one workspace.

Raw JSON units: Lab 11 stores volume in m^3 and adsorbent mass in kg: 0.001 m^3=1 L and 0.010 kg=10 g. Its pressure and isotherm parameters retain the labeled kPa basis. Workspace control strings retain the original display units.

If a result is unavailable

Read the reported reason. The vessel root must lie inside the declared pressure range and model window. Expand a range only when supported by the model/property source, not merely to force an answer. Absolute and excess uptake have different balances; do not substitute excess adsorption without its correction. No kinetics or energy balance is solved.

H1-H4 on worksheet p. 6 are optional. Their 20-minute allocation is replaced by the questions-first opening. Skip to slide 39 for the required form: the existing anonymous Session 4 exit ticket and feedback; include numerical evidence from both Labs and actual study time. Submit once, wait for “Response received,” then verify the receipt on slide 40. No separate attendance identity is collected.

Return to the complete lesson route